

CAR - Credit Assessment & Risk

Loan Default Prediction using Machine Learning

Project Explanation Guide for Beginners

Dataset	Loan_default.csv (255,347 loans, 18 columns)
Algorithms	Logistic Regression, Decision Tree, Random Forest
Metrics	Accuracy, R2 Score, MAE, RMSE
Tools	Python, Pandas, Scikit-learn, Matplotlib, Seaborn, Streamlit
Best model	Logistic Regression (88.52% accuracy)

1. Project Overview

Banks give loans to thousands of people, but some borrowers fail to repay. This is called a **loan default**. If a bank can predict who is likely to default, it can reject risky loans, charge a higher interest rate, or ask for a co-signer.

CAR (Credit Assessment & Risk) is a machine learning project that learns from past loan data and predicts whether a new borrower will **repay (0)** or **default (1)**. This is a **binary classification** problem.

The Problem Statement

Given details about a borrower (age, income, credit score, loan amount, etc.), predict whether the loan will default.

Project Goals

- Clean the raw data and make it ready for analysis.
- Visualize the data to find which factors affect default.
- Train 3 beginner-friendly algorithms and compare them with Accuracy, R2, MAE and RMSE (accuracy kept between 80% and 90%).
- Build a dashboard web app with KPIs, graphs, model comparison and a live prediction form.

Project Workflow

1. Data Cleaning notebook 1	**2. Visualization** notebook 2	**3. ML Models** notebook 3	**4. Dashboard App** CAR.py

2. Project Files

File	Purpose
1_Data_Cleaning.ipynb	Loads the data, checks missing values, duplicates and outliers, drops LoanID, saves Loan_default_cleaned.csv.
2_Data_Visualization.ipynb	Draws charts (distributions, boxplots, default rate by group, correlation heatmap) to understand the data.
3_ML_Model.ipynb	Encodes text columns, splits and scales the data, trains 3 models, calculates metrics, saves the best model.
CAR.py	Streamlit dashboard app: Dashboard, Model Comparison, Predict and About tabs.
logo.png	Logo shown in the app header, sidebar and browser tab.
requirements.txt	List of Python libraries to install with pip.

3. The Dataset

The dataset has **255,347 loans** and **18 columns**. Each row is one loan. The target column is **Default**.

Column	Meaning	Type
Age	Age of the borrower (18-69)	Number
Income	Yearly income	Number
LoanAmount	Amount of money borrowed	Number
CreditScore	Credit score (300-850); higher is better	Number
MonthsEmployed	How many months the borrower has worked	Number
NumCreditLines	Number of open credit lines (1-4)	Number
InterestRate	Interest rate of the loan (%)	Number
LoanTerm	Loan duration in months (12-60)	Number
DTIRatio	Debt-to-Income ratio (0.1-0.9)	Number
Education	High School, Bachelor's, Master's, PhD	Text
EmploymentType	Full-time, Part-time, Self-employed, Unemployed	Text
MaritalStatus	Single, Married, Divorced	Text
HasMortgage / HasDependents / HasCoSigner	Yes or No	Text
LoanPurpose	Auto, Business, Education, Home, Other	Text
LoanID	Unique ID (removed, not useful)	Text
Default	**Target: 0 = repaid, 1 = defaulted**	Number

Class imbalance: only about 11.6% of loans default (29,653 loans) while 88.4% are repaid. This matters later because it explains why model accuracy is naturally around 88-89%.

4. Step 1: Data Cleaning (notebook 1)

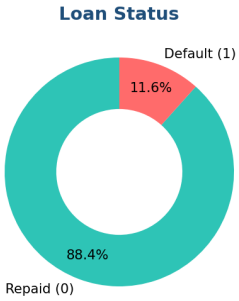
Raw data is often messy. Cleaning means fixing problems before we use the data. We did these steps:

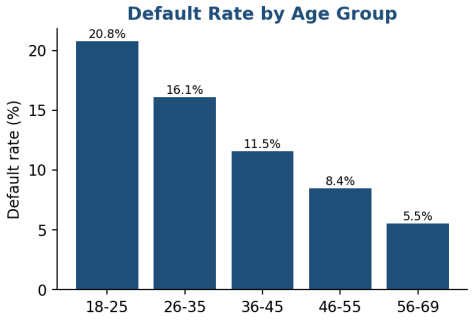
Step	What we did	Result
Load data	pd.read_csv()	255,347 rows, 18 columns
Missing values	df.isnull().sum()	0 missing values
Duplicates	df.duplicated().sum()	0 duplicates
Drop useless column	Removed LoanID (only an ID)	17 columns left
Text cleanup	Removed extra spaces in text columns	Clean categories
Outliers (IQR method)	Checked 9 numeric columns	No outliers, values are realistic
Save	Loan_default_cleaned.csv	Used by notebooks 2, 3 and CAR.py

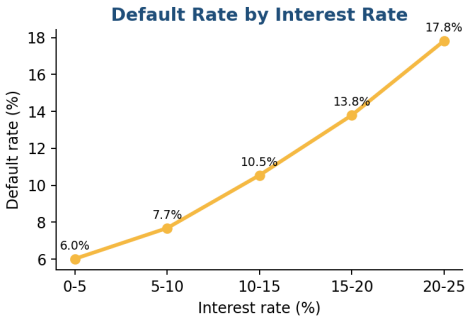
IQR method (simple idea): a value is an outlier if it is below Q1 - 1.5 x IQR or above Q3 + 1.5 x IQR, where IQR = Q3 - Q1.

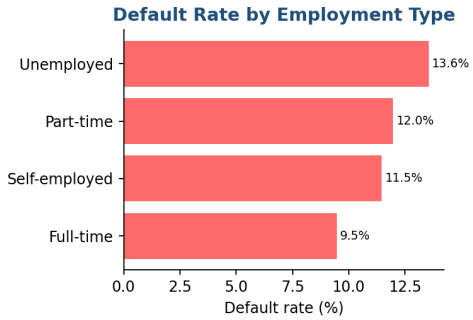
5. Step 2: Data Visualization (notebook 2)

Charts help us see patterns that are hard to notice in numbers. The main findings are:









- **Age:** younger borrowers default more (20.8% for age 18-25 vs 5.5% for age 56-69).
- **Interest rate:** the higher the interest rate, the higher the default rate (6.0% to 17.8%).
- **Employment:** unemployed borrowers default most (13.6%); full-time workers least (9.5%).
- **Co-signer, credit score, income:** no co-signer (12.9% vs 10.4%), lower credit score and lower income all go with more defaults.

6. Step 3: Machine Learning Models (notebook 3)

Preparing the data

- **Encoding:** models understand only numbers. Yes/No became 1/0, Education became 0-3 (it has an order), and EmploymentType, MaritalStatus and LoanPurpose used One-Hot Encoding (get_dummies). This gave 22 input columns.
- **Train-test split:** 80% training (204,277 rows) and 20% testing (51,070 rows), with stratify so both parts keep the same default ratio.
- **Scaling:** StandardScaler puts all columns on a similar scale so models learn better.

The 3 algorithms

Algorithm	How it works (simple)	Settings used
Logistic Regression	Draws a smooth boundary between repaid and default using a weighted sum of the inputs and gives a probability. Simple and fast.	max_iter = 1000
Decision Tree	Asks a series of yes/no questions (for example: is the interest rate above 15%?) like a flowchart until it reaches a decision.	max_depth = 5 (keeps it simple, avoids overfitting)
Random Forest	Builds many decision trees and lets them vote. Usually more stable than a single tree.	100 trees, max_depth = 8

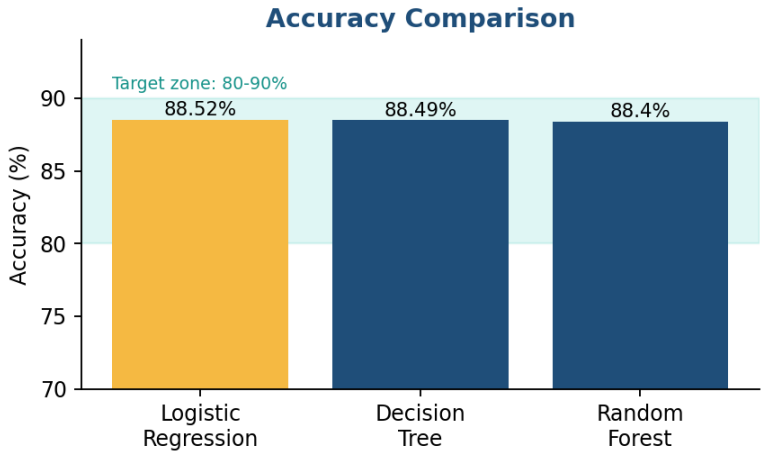
The 4 metrics

Metric	Meaning	Better when
Accuracy	Percentage of loans predicted correctly.	Higher
R2 Score	How much of the variation in the target the model explains.	Higher
MAE	Mean Absolute Error: average size of the prediction mistakes.	Lower
RMSE	Root Mean Squared Error: like MAE but punishes big mistakes more.	Lower

Note: R2, MAE and RMSE are normally used for regression. Here they are calculated on the 0/1 predictions, so they show how far predictions are from the real labels. A slightly **negative R2** is normal for this kind of data.

7. Results: Which Algorithm is Best?

Algorithm	Accuracy	R2 Score	MAE	RMSE
Logistic Regression (Best)	**88.52%**	-0.1180	0.1148	0.3388
Decision Tree	88.49%	-0.1215	0.1151	0.3393
Random Forest	88.40%	-0.1304	0.1160	0.3406



How the best model is selected: keep only models whose accuracy is between 80% and 90%; the highest accuracy wins; if tied, the lowest RMSE wins. **Logistic Regression** wins, with the highest accuracy, highest R2 and lowest MAE and RMSE.

Important to understand: the three models are extremely close (difference of 0.12%). Because 88.4% of loans are repaid, a model that always says "will repay" would already score 88.4%. So the accuracy range of 80-90% is natural for this dataset, and the models are mostly valuable for their **default probability** output, which is used in the app to show LOW, MEDIUM or HIGH risk.

8. Step 4: The Dashboard App (CAR.py)

CAR.py is a web app built with **Streamlit**. It has a logo, a sidebar with filters, and four tabs:

Tab	What it shows
Dashboard	6 KPI cards (Total Loans, Default Rate, Avg Loan Amount, Avg Income, Avg Credit Score, Avg Interest Rate) and 6 graphs (loan status donut, default rate by age group, employment type, credit score, interest rate and loan purpose).
Model Comparison	Best-algorithm banner, 5 KPI cards (algorithm, accuracy, R2, MAE, RMSE), a score table of all 3 models and 2 comparison charts.
Predict Default	A form for borrower details. Choose an algorithm and get the risk level (Low, Medium, High), default probability, decision, and what all 3 algorithms say.
About	Project description, tools used and how to use the app.

How a prediction works

- The user enters the borrower details in the form.
- The inputs are encoded like the training data (22 columns) and scaled with the same scaler used in training.
- The selected model returns a decision (0 or 1) and a default probability.
- Risk level: below 15% = LOW, 15% to 50% = MEDIUM, 50% or more = HIGH.

9. How to Run the Project

Step 1: Put Loan_default.csv, the 3 notebooks, CAR.py and logo.png in one folder.

Step 2: Install the libraries:

```
pip install -r requirements.txt
```

Step 3: Open and run the notebooks in order (Run All): 1_Data_Cleaning, then 2_Data_Visualization, then 3_ML_Model.

Step 4: Start the app from the terminal:

```
streamlit run CAR.py
```

The app opens in your browser at http://localhost:8501. On the first start it trains the 3 models (about 30 seconds); after that it is cached and fast.

10. Conclusion and Future Work

We built a complete beginner-friendly machine learning project: data cleaning, visualization, three models with four evaluation metrics, and an interactive dashboard. **Logistic Regression** was the best algorithm with 88.52% accuracy, and the data showed that younger age, high interest rate, unemployment and no co-signer are linked to higher default risk.

Future ideas: handle class imbalance (class weights or SMOTE) and judge models with precision, recall and ROC-AUC; try Gradient Boosting or XGBoost; deploy the app online.